

11th grade

Learning evidence 2.1: Confidence Interval Planning

Name: _____

Evaluated criteria

This exam evaluates only the criteria C1–C5. Each criterion is evaluated across the three problems below. A criterion is passed if it is correctly activated in at least two of the three problems. The overall exam result is binary (Passed/Not passed).

- C1 Compute the sample mean and sample standard deviation.
- C2 Distinguish between sample statistics and population parameters.
- C3 Explain why interval estimation is preferred over a single point estimate.
- C5 Construct a 95 % confidence interval using known population standard deviation.
- C4 Interpret the meaning of a 95 % confidence interval in context.

Problem 1: A city budgeting office wants to estimate the average weekly transit subsidy per rider for two pilot zones. Zone A recorded the following sample of weekly subsidy amounts (USD):

- Zone A sample ($n_A = 6$): 18, 22, 19, 21, 20, 17.
- Zone B sample ($n_B = 8$): 24, 26, 23, 25, 27, 22, 24, 26.

An annual audit provides the population standard deviations for the zones: $\sigma_A = 3,5$ and $\sigma_B = 4,0$. For classroom purposes, suppose the true population mean weekly subsidy is $\mu_A = 20$ dollars in Zone A and $\mu_B = 25$ dollars in Zone B. Construct and interpret a 95 % confidence interval for the population mean weekly subsidy in each zone.

Problem 2: A regional warehouse studies the number of orders processed per hour to plan staffing. A random sample of 60 hours is summarized in grouped form below. The operations team also reports a known population standard deviation of $\sigma = 6,2$ orders per hour. For academic purposes only, assume the true population mean order rate is $\mu = 22$ orders per hour. The grouped data pairs (orders per hour, frequency) are: $14 \rightarrow 8$, $18 \rightarrow 16$, $22 \rightarrow 20$, $26 \rightarrow 10$, and $30 \rightarrow 6$. Construct and interpret a 95 % confidence interval for the population mean orders per hour.

Problem 3: A logistics firm monitors the time (in minutes) required to load containers at a port. A sample of 80 loading times is grouped into class intervals below. The firm has a historical estimate of the population standard deviation, $\sigma = 5,8$ minutes. For classroom reference, suppose the true population mean loading time is $\mu = 32$ minutes. The grouped intervals and frequencies are: 20–24 minutes (12), 25–29 minutes (18), 30–34 minutes (20), 35–39 minutes (17), and 40–44 minutes (13). Construct and interpret a 95 % confidence interval for the population mean loading time.